Baxter Harrington

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EDUCATION

Washington University in St. Louis | McKelvey School of Engineering **Class of 2028**
B.S. in Mechanical Engineering | Minor in Human-Computer Interaction
GPA: 3.88/4.00
Coursework: Solid Mechanics, Fluid Mechanics, Differential Equations, Machine Elements, Thermodynamics, Materials Science, Intro to Computer Science (Java), Web Development

PROJECTS & EXPERIENCE

SO-101 Robotic Arm Project **Summer 2026 - Present**
- Built and calibrated a LeRobot SO-101 leader-follower robotic arm system from open-source hardware designs.
- Collected an imitation-learning demonstration dataset via teleoperation and trained Action Chunking with Transformers (ACT) policies for a pick-and-place task.
- Diagnosed the dominant failure mode (gripper closing behind the block, 22/50 trials) as a depth-perception problem caused by a shallow camera angle; designed and 3D-printed an overhead camera mount and retrained, raising success from 32% to 68% on comparable placements (n=50 / n=41) and cutting that failure to 4 occurrences.
- Developing custom environments in Gymnasium for SO-101 block manipulation to train reinforcement learning policies in simulation for sim-to-real transfer to the SO-101.

University Rover Challenge | Mechanical Engineer → Science Module Engineering Lead **Fall 2025 - Present**
- Leading a 6-person design team toward the May 2027 competition in Southern Utah.
- Designing a soil-collection and testing module for a Mars rover from the ground up, integrating robotic mechanisms and chemical tests for in-situ analysis.
- Prototyped the sample collection system during the 2025-2026 academic year, using SolidWorks and 3D printing to develop proof of concepts for both surface and deep-soil extraction mechanisms.

Konica Minolta Business Solutions | Information Technology Intern **Summer 2026**
- Facilitated discovery interviews across 3 business units to surface jobs-to-be-done for an agentic AI (Salesforce Agentforce) rollout.
- Synthesized findings into 4 shared pain points and matching agent designs.
- Prototyped all 4 agent concepts as interactive HTML mockups using Claude Design, then returned to the same interview participants for 30-minute validation sessions.

Addison Central School District | Technology Support Intern **Summer 2025**
- Configured and deployed 500 new computers and refurbished 500 computers for redistribution, using imaging tools, BIOS configuration, and diagnostic utilities.

SKILLS

Robotics & Embodied AI: LeRobot, imitation learning (ACT), teleoperation data collection, Gymnasium
Mechanical Engineering: SolidWorks (CAD, simulation/FEA), 3D printing/rapid prototyping, machining (mill, CNC, lathe)
Experience Design & Research: discovery interviews, usability/validation testing, Figma, Claude Design
Programming: Python, Java, JavaScript, HTML/CSS, MATLAB, Git

LEADERSHIP & ACTIVITIES

KEYS Volunteer Piano Lessons | Treasurer **Fall 2025 - Present**
- Managing finances for educational materials and recitals of a club providing free piano lessons to kids in St. Louis.

Club Lacrosse | Treasurer **Fall 2025 - Spring 2026**
- Managed finances for travel, games, and equipment of a 30-person team.

INTERESTS

Cross country skiing, Formula 1, piano, Rubik's Cube (personal best 14 seconds)